



Review

Vaccination of special populations: Protecting the vulnerable



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ABSTRACT

One of the strategic objectives of the 2011–2020 Global Vaccine Action Plan is for the benefits of immunisation to be equitably extended to all people. This approach encompasses special groups at increased risk of vaccine-preventable diseases, such as preterm infants and pregnant women, as well as those with chronic and immune-compromising medical conditions or at increased risk of disease due to immunosenescence. Despite demonstrations of effectiveness and safety, vaccine uptake in these special groups is frequently lower than expected, even in developed countries with vaccination strategies in place. For example, uptake of the influenza vaccine in pregnancy rarely exceeds 50% in developed countries and, although data are scarce, it appears that only half of preterm infants are up-to-date with routine paediatric vaccinations. Many people with chronic medical conditions or who are immunocompromised due to disease or aging are also under-vaccinated. In the US, coverage among people aged 65 years or older was 67% for the influenza vaccine in the 2014–2015 season and 55–60% for tetanus and pneumococcal vaccines in 2013, while the coverage rate for herpes zoster vaccination among those aged 60 years or older was only 24%. In most other countries, rates are far lower. Reasons for under-vaccination of special groups include fear of adverse outcomes or illness caused by the vaccine, the inconvenience (and in some settings, cost) of vaccination and lack of awareness of the need for vaccination or national recommendations. There is also evidence that healthcare providers' attitudes towards vaccination are among the most important influences on the decision to vaccinate. It is clear that physicians' adherence to recommendations needs to be improved, particularly where patients receive care from multiple subspecialists and receive little or no care from primary care providers.

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Abbreviations: ACIP, Advisory Committee on Immunisation Practices; CDC, Centers for Disease Control and Prevention; GVAP, Global Vaccine Action Plan; Hib, *Haemophilus influenzae* type b; HIV, human immunodeficiency virus; HPV, human papillomavirus; NICU, neonatal intensive care unit; PCV, pneumococcal conjugate vaccine; Td, tetanus-diphtheria vaccine; Tdap, Td-acellular pertussis vaccine; WHO, World Health Organization.

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1. Introduction

One of the strategic objectives of the 2011–2020 Global Vaccine Action Plan (GVAP) is for the benefits of immunisation to be equitably extended to all people [1]. This approach encompasses special groups at increased risk of vaccine-preventable disease, such as preterm infants and pregnant women, as well as those with chronic and immune-compromising medical conditions. The increased risk of disease due to immunosenescence with advancing age must also be taken into consideration in the development of vaccination strategies that cover all ages and health conditions. Special populations are often under-vaccinated for various reasons, including lack of awareness of vaccine-preventable diseases and uncertainty or misconceptions about the safety and efficacy of vaccination among patients, parents and healthcare providers, as well as cost and the inability of healthcare systems to ensure such patients receive recommended vaccines. In this review, we discuss vaccination strategies in the context of special situations or conditions that increase vulnerability to disease, focusing mainly on recommendations of the World Health Organization (WHO) and the US Centers for Disease Control and Prevention (CDC) Advisory Committee on Immunisation Practices (ACIP). Rationales that support recommended vaccination strategies, as well as barriers to the immunisation of special populations, are also considered.

2. Vaccination in pregnancy

The physiological changes associated with pregnancy can lead to an elevated risk for severe disease [2] (Box 1). Influenza infection during pregnancy also has risks for the foetus, including premature birth, reduced birth weight and an elevated risk of death [3]. The 2009 influenza pandemic was a timely reminder of the risks: surveillance data from the US estimated that influenza infection during pregnancy was associated with a sevenfold increased risk of hospitalisation and fourfold increased risk of admission to intensive care units or death when compared to infected non-

pregnant women [4]. During the winter of 2009 in Australia and New Zealand, the proportion of patients with influenza who were pregnant and admitted to an intensive care unit was nine times higher than the corresponding proportion in the general population [5]. Advisory groups throughout the world have therefore recommended influenza vaccination for pregnant women, with the WHO identifying them as a priority group [6,7]. The positive impact of maternal influenza vaccination is not limited to developed countries [7]. For example, a randomised controlled trial with trivalent inactivated influenza vaccine in Bangladesh suggested maternal vaccination had effectiveness of 63% against confirmed influenza in infants, with reduction of influenza-like illness in infants and mothers by 29% and 36%, respectively [8]. Other study data suggest influenza vaccination was associated with a significant decline in the risk of infants being small for gestational age [9]. However, for countries with very constrained health budgets, influenza prevention is rarely a priority [10].

Vaccination against tetanus and pertussis is also widely recommended (Table 1). Neonatal tetanus has a very high case fatality rate and although improved hygiene during birth reduces the risk of maternal and neonatal tetanus, maternal vaccination has been critical in reducing the incidence, especially where good quality perinatal care is not available and home births are frequent [11]. Effectiveness trials demonstrated reductions in neonatal tetanus of 80%, with declines in mortality of up to 98% after maternal receipt of two or three vaccine doses [12]. Between 1988 and 2013, the estimated number of deaths due to tetanus among neonates fell from 787,000 to 49,000 globally, which was attributed mostly to maternal tetanus vaccination [13]. A key component of this was the establishment of supplementary immunisation activities, which ensured that as many pregnant women as possible had access to the vaccine [11].

Pertussis (whooping cough) remains endemic in much of the world. Infants are especially vulnerable in the early months of life before vaccination, once protective maternal antibody levels have waned [14]. Since vaccination of the family to prevent pertussis transmission (‘cocooning’) has shown little evidence of effectiveness, health authorities now increasingly advise against the

Box 1
Vaccination in pregnancy.

Population	• Pregnant women
Reasons for increased risk of disease	• Altered state of immune responsiveness due to adaptive immune responses by mother to tolerate foetus and by foetus to tolerate mother
Diseases most commonly targeted by vaccination	• Influenza
	• Tetanus
	• Pertussis
Reported vaccine uptake rates ^a	• Influenza vaccine uptake ~50% or less
Common barriers to vaccination	• Failure of healthcare provider to recommend vaccination
	• Lack of knowledge of vaccine safety and effectiveness; fear of adverse pregnancy outcomes
	• Perceived susceptibility to disease
	• Lack of awareness of national recommendations
	• Inconvenience
Unmet vaccination needs	• Vaccination against other leading causes of infection during early infancy
	• Group B Streptococcus
	• Respiratory syncytial virus

^a In US and England.

Table 1

Vaccines recommended by the World Health Organization (WHO) and US Centers for Disease Control and Prevention Advisory Committee on Immunisation Practices (ACIP) for specific groups: pregnant women, preterm infants, adolescents with chronic medical conditions and older adults.

Special population	Vaccine	Recommendation (advisory body)
Pregnant women ^a	Seasonal inactivated influenza	One dose at any stage of pregnancy (WHO, ACIP)
	T or Td Tdap	All women giving birth and their newborn babies should be protected against tetanus (WHO) One dose in countries or settings with high or increasing infant morbidity/mortality from pertussis (WHO) One dose in each pregnancy (ACIP)
Preterm infants	All recommended paediatric vaccines	Preterm infants who are otherwise healthy should be immunised according to the same vaccination schedule as used for full-term infants (WHO, ACIP)
	Hepatitis B	Birth doses given to infants <2 kg should not be counted towards the primary vaccination series because of the possibility of poor response (WHO, ACIP)
	PCV	Booster dose in second year of life if received three primary doses before 12 months of age (WHO)
Adolescents with chronic medical conditions	HPV	As for general population (WHO/ACIP) except: Extended duration of catch-up vaccination for immunocompromised males (ACIP) Increased dosing requirements for immunocompromised females <15 years (WHO)
	Td/Tdap	As for general population (WHO/ACIP) except: Consider shorter dosing interval for Tdap for adolescents with high risk conditions (ACIP)
	Meningococcal conjugate vaccine	As for general population (WHO/ACIP) except: For high-risk conditions (asplenia, complement component deficiency, HIV), two primary doses and (if asplenia or complement component deficiency) booster dose every 5 years (ACIP)
	Seasonal inactivated influenza	One dose annually (WHO, ACIP)
	Pneumococcal	Catch-up with PCV if high-risk condition, e.g., CSF leak, cochlear implant, asplenia, immunocompromised (ACIP) One dose of pneumococcal polysaccharide vaccine if high-risk condition and additional dose after 5 years if asplenia or immunocompromised (ACIP)
Older adults ^a	Seasonal inactivated influenza	≥ 65 years of age: one dose annually (WHO, ACIP) ^b
	Tdap/Td	≥ 65 years of age: substitute one-time dose of Tdap for Td booster, then boost with Td every 10 years (ACIP) ^b
	Herpes zoster	≥ 60 years of age: one dose (ACIP)
	Pneumococcal	≥ 65 years of age: one dose of PCV and/or pneumococcal polysaccharide vaccine, depending on vaccination history (ACIP)

Abbreviations: T, tetanus toxoid; d, reduced diphtheria toxoid; ap, reduced acellular pertussis; CSF, cerebrospinal fluid; HIV, human immunodeficiency virus; HPV, human papillomavirus; MenC, meningococcal group C; PCV, pneumococcal conjugate vaccine.

^a Other vaccines may be recommended based on risk factors and special circumstances, including hepatitis A, hepatitis B and meningococcal vaccines.

^b Also recommended for younger adults.

practice in favour of maternal immunisation [15,16]. In 2011–2012, some countries began to recommend maternal vaccination with the tetanus-diphtheria-acellular pertussis (Tdap) vaccine, to protect against pertussis infection in early infancy [17]. Following an outbreak of pertussis in England in 2012, a study of 26,684 women (64% of whom were vaccinated) indicated that the vaccine reduced the risk of pertussis in infants by 91% [18].

While the benefits of maternal vaccination with recommended vaccines are clear, safety concerns have meant that pregnant women have traditionally been excluded from clinical trials [19,20]. The data for both tetanus and influenza vaccination therefore rest on decades of experience from very large retrospective cohorts: more than 170 million women of childbearing age have been vaccinated against tetanus [13], many of them during pregnancy, while influenza vaccination is given to millions of pregnant women annually. Among the estimated 2 million pregnant women immunised in the US during the 2000–2003 influenza seasons, only 20 adverse events were reported to the CDC Vaccine Adverse Event Reporting System, of which 17 were minor injection-site or systemic reactions [14]. There are fewer data on Td or Tdap uptake, due to the recent introduction of the vaccine for use during pregnancy and the fact that most of the global elimination campaigns have been carried out with tetanus toxoid vaccine. Nonetheless, a US study of 26,229 women vaccinated with Tdap matched with 97,265 unvaccinated women found no evidence of an increased risk of adverse obstetric events apart from a small increased risk of chorioamnionitis [21]. Similarly, a UK study of maternal pertussis immunisation in 20,074 women found no increased risk for an extensive predefined list of adverse events related to pregnancy [22].

Despite such evidence, uptake of recommended vaccines among pregnant women has tended to be lower than expected [23,24]. For example, in the 2013–14 influenza season, 52% of women in the US were vaccinated just before or during pregnancy [23], even though the vaccine is recommended for all pregnant women. Influenza and pertussis vaccine uptake in pregnancy in England is around 42% and 60%, respectively [25,26]. The UK has among the highest coverage rates globally [27], indicating the scale of the problem.

Common reasons for not receiving vaccination include fear of adverse pregnancy outcomes, the inconvenience (and in some settings, cost) of vaccination, lack of awareness of national recommendations and failure of the healthcare provider to recommend vaccination [19,23,24]. In low-income countries, pregnant women are most likely to cite access issues as barriers to vaccination [19]. In countries where access is not a barrier, the healthcare providers' recommendation and offer of vaccination appears to be the most influential factor on whether a pregnant woman is vaccinated or not. As discussed in an accompanying paper [28], healthcare providers have a central role in enhancing knowledge of vaccine safety in all patient groups, including pregnant women.

3. Vaccination of preterm infants

Infection tends to have more serious consequences in preterm than in full-term infants, mainly because of immaturity of the immune system [29] (Box 2). Consequently, preterm and low birth weight infants have a higher risk of vaccine-preventable diseases, including those caused by pertussis, *Streptococcus pneumoniae* and rotavirus [30–32].

Box 2

Vaccination of preterm infants.

Population	• Preterm infants
Reasons for increased risk of disease	• Immaturity of the immune system
Diseases most commonly targeted by vaccination	• If clinically stable and not in intensive care: same diseases as targeted for general population of infants
	• If hospitalised: diphtheria, tetanus, pertussis, hepatitis B, <i>Haemophilus influenzae</i> type b, poliovirus, pneumococcal
Reported vaccine uptake rates ^a	• 40–50% not up to date with routine immunisations
Common barriers to vaccination	• Physicians' inadequate knowledge of efficacy and safety of vaccines in this group
	• Reluctance to vaccinate during hospitalisation
Unmet vaccination needs	• Vaccination against
	• Group B Streptococcus
	• Respiratory syncytial virus
	• Rotavirus when in neonatal intensive care unit

^a In US and Australia.

It is generally recommended that preterm infants who are otherwise healthy are immunised according to the vaccination schedule used for full-term infants (Table 1). To ensure early protection, preterm infants should be vaccinated according to their chronological rather than corrected gestational age and regardless of birth weight [29], although it may be appropriate to administer additional vaccine doses to preterm or extremely low birth weight infants who produce suboptimal vaccine responses. For hepatitis B vaccination, guidelines recommend that birth doses given to infants under 2 kg should not be counted towards the primary vaccination series because of the possibility of a poor response [6,33]. Similarly, since a pneumococcal conjugate vaccine (PCV) booster dose is not administered in all countries [34], WHO guidelines specifically recommend a booster dose in the second year of life for preterm infants who received three primary doses before 12 months of age [6] (Table 1).

Preterm infants are usually adequately protected by routine paediatric vaccines, with immune responses lower than in full-term infants but high enough to provide protection [29]. In a study of the 10-valent PCV (*Synflorix*TM, GSK Vaccines, Belgium), prevaccination antibody concentrations tended to be higher for most vaccine pneumococcal serotypes in the term versus preterm groups [35]. However, after three-dose primary vaccination at 2–4–6 months of age and a booster dose at 16–18 months, at least 98% of children in all term and preterm groups reached the protective antibody threshold for each vaccine pneumococcal serotype.

Despite demonstrations of protective immune responses, routine immunisation of premature infants is often delayed because of physicians' inadequate knowledge of the immunogenicity or safety of vaccines. For example, in an audit undertaken in Melbourne, Australia, the 6-month Tdap and poliovirus vaccine doses were delayed by more than one month in 43% of preterm infants and uptake of additional recommended vaccine doses was 19–35% [36]. An education programme for healthcare professionals regarding guidelines was introduced as well as immunisation 'stickers' for health records of preterm infants [37]. In an audit conducted four years later, 96% of preterm-born children were up to date with routine immunisations at 12 months of age and receipt of additional recommended hepatitis B and influenza vaccines increased by more than twofold. Also, vaccination of preterm infants may be discouraged during hospitalisation, as has been the case for rotavirus vaccination of infants in neonatal intensive care units (NICUs) [38]. Since the ACIP recommends immunisation at the time of discharge if the child is clinically stable and aged 42–104 days, preterm infants who remain in the NICU beyond the upper age limit are ineligible for rotavirus vaccination [39]. However, rotavirus vaccination within the NICU has been shown to be well tolerated, with no evidence of transmission to contacts [38,40], indicating that the risk of severe disease in unvaccinated infants outweighs the risk of transmission during hospitalisation.

Despite this, a study of six NICUs in California found that only 51% of discharged infants were up-to-date for vaccines recommended for use in the NICU (Tdap, hepatitis B, *Haemophilus influenzae* type b [Hib] and poliovirus vaccines) [41]. These examples highlight the need for further demonstrations of vaccination benefits for preterm infants and for alternative approaches to ensure they receive age-appropriate recommended vaccines.

4. Vaccination of individuals with immune-compromising diseases or other chronic conditions

The number of patients who are immunocompromised or live with chronic conditions is increasing [42]. Immunocompromised patients include those with primary (hereditary or genetic) immunodeficiency disorders or secondary immunodeficiencies that are generally acquired and occur due to a disease process or its therapy (Table 2). This includes human immunodeficiency virus (HIV) infection, cancer, transplantation, asplenia or sickle cell disease and autoimmune inflammatory diseases treated with immunosuppressive medications (corticosteroid therapy, immunomodulatory medications or biological agents) [43–46]. Other populations with chronic conditions may be generally immunocompetent but at elevated risk for certain vaccine-preventable infections. Examples include individuals with diabetes [47], cystic fibrosis [48] or anatomic barrier defects, such as those with cochlear implants [43]. In cystic fibrosis, it is especially important to avoid diseases that could cause pulmonary deterioration, such as pneumonia and influenza [48], while bacterial meningitis is a particular concern for cochlear implant recipients [49].

Among the immunocompromised population, the severity of immunosuppression varies depending on the condition and treatment drugs used. Thus HIV-infected patients with a CD4 T-cell count under 200 cells/mm³ (for adults), advanced Hodgkin's disease patients and haematopoietic stem cell transplant recipients rank amongst the most immunocompromised groups [42,43], whereas those treated with biological agents for multiple sclerosis, inflammatory bowel disease and rheumatoid arthritis are usually moderately compromised [44,45,50]. The type of immune deficiency also varies according to underlying disease, e.g., leukaemic patients may suffer from the effects of neutropenia, HIV/AIDS patients from T-cell deficiencies, while those with some congenital immunodeficiencies from B-cell defects.

These factors influence the infections to which immunocompromised patients are predisposed and choice of immunisation strategy. For example, reactivation of herpesviruses is a classic complication of solid organ and haematopoietic stem cell transplantation [51]. Live attenuated vaccines cannot be used in severely immunocompromised patients because of the risk of inducing disease but may be safe in mild or moderately immuno-

Table 2

Effectiveness of vaccination in persons with primary (hereditary or genetic) and secondary (acquired) immunodeficiencies (adapted from ACIP General Recommendations on Immunisation [46]).

Immunodeficiency	Specific immunodeficiency/condition	Effectiveness of vaccination ^a
<i>Primary</i>		
B cell (humoral)	Severe antibody deficiencies (e.g., X-linked agammaglobulinaemia and common variable immunodeficiency) Less severe antibody deficiencies (e.g., selective IgA deficiency and IgG subclass deficiency)	The effectiveness of any vaccine is uncertain if it depends only on the humoral response (e.g., PPSV or MPSV4) IGIV interferes with the immune response to measles vaccine and possibly varicella vaccine All vaccines likely effective; immune response might be attenuated
T cell (cell-mediated and humoral)	Complete defects (e.g., severe combined immunodeficiency disease, complete DiGeorge syndrome) Partial defects (e.g., most patients with DiGeorge syndrome, Wiskott-Aldrich syndrome, ataxia-telangiectasia)	Vaccines might be ineffective Effectiveness of any vaccine depends on degree of immune suppression
Complement	Persistent complement, properdin or factor B deficiency	All routine vaccines likely effective
Phagocytic function	Chronic granulomatous disease, leukocyte adhesion defect, myeloperoxidase deficiency	All inactivated vaccines likely effective Live viral vaccines likely effective
<i>Secondary</i>		
	HIV infected	MMR, varicella, rotavirus and all inactivated vaccines, including inactivated influenza, might be effective Effectiveness of any vaccine depends on degree of immune suppression
	Malignant neoplasm, transplantation, immunosuppressive or radiation therapy Asplenia or sickle cell diseases Chronic renal disease	All routine vaccines likely effective All routine vaccines likely effective

Abbreviations: HIV, human immunodeficiency virus; Ig, immunoglobulin; IGIV, immune globulin intravenous; MMR, measles, mumps and rubella vaccine; MPSV4, quadrivalent meningococcal polysaccharide vaccine; PPSV, pneumococcal polysaccharide vaccine.

^a See Table 3 for the Infectious Diseases Society of America list of recommended and contraindicated vaccines.

compromised patients. Inactivated and subunit vaccines are the best alternative although in some cases live attenuated vaccines can be administered up to a month before patients are predicted to become immunocompromised [43]. There is much ongoing research on vaccination of this group, particularly those with severe T-cell-mediated immunodeficiencies, partly guided by recent successes with immunotherapy [52,53]. This includes vaccines against diseases caused by herpesviruses and polyomaviruses, especially in transplant patients, tuberculosis in HIV-induced immunodeficiency and for herpes zoster and fungi in haematopoietic stem cell transplantation.

The Infectious Diseases Society of America clinical practice guideline for vaccination of immunocompromised patients is summarised in Table 3 [43]. This demonstrates the range of approaches required for the wide variety of conditions involved. We briefly outline issues involved with two conditions: asplenia, a condition with a high risk of infectious disease and clear recommendations for vaccination, and diabetes, an at-risk condition that is frequently under-vaccinated.

Asplenia results from surgical removal of the spleen or atrophy, for example following trauma, splenic artery thrombosis, hypersplenism (e.g., after schistosomiasis) or recurrent infarctions (e.g., in sickle cell disease). Since encapsulated bacteria are primarily removed by the spleen, hypo- or asplenic patients are susceptible to severe, often fulminant, infections with encapsulated bacteria, particularly *S. pneumoniae*, but also *Neisseria meningitidis* and Hib [54,55]. Consequently, current guidelines for hypo- and asplenic patients emphasise that, in addition to routine vaccination, this group should be immunised against pneumococcal, meningococcal and Hib infection as well as against influenza linked to the associated risk of secondary bacterial infection [54].

In diabetes, immunodeficiency due to genetic and metabolic abnormalities and other factors (age, renal disease and cardiovascular disease) may account for the increased severity of *S. pneumoniae* and influenza virus infections in this patient group compared with non-diabetic individuals [56]. Guidelines in various countries thus recommend influenza and pneumococcal vaccines for this

patient group [57–59]. Epidemiological data from the US suggest an increased risk for acute hepatitis B among adults with diabetes [60,61] and chronic hepatitis may have a more severe course in diabetes patients [62]. Evidence also suggests that diabetes is a risk factor for herpes zoster [63–65] and post-herpetic neuralgia [63,64,66]. In 2011, the ACIP recommended hepatitis B vaccination for previously unvaccinated adults aged 19–59 years with diabetes as soon as possible after diagnosis and that adults with diabetes aged 60 years or older are vaccinated at the discretion of the treating clinician [67]. There is currently no specific recommendation for vaccinating people with diabetes against herpes zoster.

There are few data on vaccine coverage levels for patients who are immunocompromised or live with chronic conditions with increased risks for vaccine-preventable disease. Available evidence suggests that these patient groups are often under-vaccinated, even in countries with well-functioning healthcare systems [44,45,68–70]. For example, a US study of vaccination in patients with diabetes found coverage rates of 41% for influenza and 37% for pneumococcal vaccination, while no patients had received the hepatitis B vaccine despite ACIP guidelines [70]. This may be partly because of a lack of awareness of the importance of vaccination and misperceptions about vaccine safety and immunogenicity in these groups. Patients may miss routine vaccinations because of frequent hospital admissions and school absenteeism. Also, in certain countries, such as the US, patients with chronic medical problems are likely to receive care from a specialist physician rather than a primary care provider. Vaccinations may therefore be missed if specialists assume vaccination is the responsibility of a general practitioner.

4.1. Vaccination of adolescents with chronic medical conditions

Adolescents are an important group to consider separately because they are at increased risk of several vaccine-preventable infections, particularly pertussis, human papillomavirus (HPV) and meningococcal infections [71]. This is due to various factors including waning immunity to vaccination in the case of pertussis,

Table 3
Vaccines recommended by the Infectious Diseases Society of America (IDSA) for individuals who are immunocompromised because of their condition or treatment of their condition and contraindicated vaccines (summary of 2013 IDSA clinical practice guideline [43]).

Patient population/condition	Recommended vaccines	Contraindicated vaccines
Primary immunodeficiency		
– Primary (congenital) complement deficiencies	As for general population	None
– Phagocytic cell deficiency	Inactivated vaccines, as for general population	Live vaccines for majority of patients
– Innate immune defects resulting in cytokine or cellular activation defects	Inactivated vaccines, as for general population	Live vaccines for majority of patients
– Minor antibody deficiencies	As for general population	OPV for IgA-deficient patients
– Major antibody deficiencies receiving immunoglobulin therapy or combined immunodeficiencies	Inactivated influenza vaccine can be administered if residual antibody production	Inactivated vaccines (other than influenza vaccine), live vaccines
HIV infected	As for general population except for contraindicated vaccines	Live attenuated influenza vaccine Live MMR or VAR for children with CD4 T-cell percentage <15 or adults/adolescents with CD4 T-cell count <200 cells/mm ³ Live MMRV vaccine Live HZ vaccine
Cancer	Inactivated influenza vaccine annually except those receiving anti-B-cell antibodies or intensive chemotherapy Other inactivated vaccines for immunocompetent children receiving maintenance chemotherapy PCV followed by PPSV Three months after chemotherapy or at least 6 months after receiving anti-B-cell antibodies, can administer inactivated vaccines and live VAR, MMR, MMRV vaccines as for general population	Live viral vaccines during chemotherapy
Haematopoietic stem cell transplant	Pre-transplant: As for general population if not immunosuppressed and if interval to start of conditioning regimen is ≥4 weeks for live vaccines and ≥2 weeks for inactivated vaccines. VAR for nonimmune transplant candidates if interval to start of conditioning regimen is ≥4 weeks Starting 6 months after transplant: inactivated influenza vaccine, Td-containing vaccine Starting 3–6 months after transplant: PCV/PPSV Starting 6–12 months after transplant: Hib, meningococcal conjugate, hepatitis B, IPV, HPV vaccines Starting 24 months after transplant: MMR if measles-seronegative and VAR if varicella-seronegative and no graft vs host disease or ongoing immunosuppression	Live vaccines if <4 weeks and inactivated vaccines if <2 weeks before start of conditioning regimen After transplant: live vaccines to patients with active graft vs host disease or ongoing immunosuppression
Solid organ transplant	Pre-transplant: As for general population, PCV/PPSV, hepatitis B, hepatitis A, HPV (age 11–26 years) vaccines. MMR and VAR (age 6–11 months) if not receiving immunosuppression and >4 weeks to transplant. HZ (age 50–59 years) if not severely immunocompromised and >4 weeks to transplant Starting ≥1 month after transplant: inactivated influenza vaccine if disease outbreak Starting 2–6 months after transplant: As for general population. If not administered before transplant, PCV with timing based on degree of immunosuppression 2–6 months after liver transplant: hepatitis B vaccine	No vaccines in first 2-month period post-transplant, apart from influenza vaccine if necessary MMR, VAR should generally not be administered to transplant recipients, except VAR in non-immune children who are renal or liver transplant recipients and receiving minimal or no immunosuppression
Chronic inflammatory diseases on immunosuppressive medications	As for general population: inactivated vaccines, PCV/PPSV VAR if non-immune and ≥4 weeks to immunosuppression initiation or if low-level, long-term immunosuppression HZ (age ≥60 years or 50–59 years and varicella-seropositive) if before immunosuppression initiation or low-dose immunosuppression	Live vaccines LAIV, MMR, MMRV
Asplenia or sickle cell disease	As for general population aged <2 years, including PCV If aged ≥2 years, PCV/PPSV If aged ≥5 years, Hib vaccine If aged ≥2 months, meningococcal vaccine but not quadrivalent conjugate vaccine in patients aged <2 years because of reduced antibody response to PCV when co-administered. If aged ≥55 years, revaccination with quadrivalent vaccine every 5 years	LAIV
Anatomic barrier defects at risk of vaccine-preventable infection	As for general population: Adults and children with profound deafness scheduled to receive cochlear implant, congenital dysplasias of the inner ear or persistent CSF communication with oropharynx or nasopharynx Patients with or ≥2 weeks before cochlear implant or with persistent CSF communication with oropharynx or nasopharynx: PCV/PPSV	None

Abbreviations: CSF, cerebrospinal fluid; Hib, *Haemophilus influenzae* type b vaccine; HIV, human immunodeficiency virus; HPV, human papillomavirus vaccine; HZ, herpes zoster vaccine; Ig, immunoglobulin; IPV, inactivated poliovirus vaccine; LAIV, live attenuated influenza vaccine; MMR, measles, mumps and rubella vaccine; MMRV, MMR-varicella vaccine; OPV, oral poliovirus vaccine; PCV, pneumococcal conjugate vaccine; PPSV, pneumococcal polysaccharide vaccine; Td, tetanus toxoid, reduced diphtheria toxoid; VAR, varicella vaccine.

and exposure risks such as an elevated risk of sexually-transmitted infections. For the latter, vaccination is available against HPV and there are several promising vaccine candidates against sexually-transmitted herpes and chlamydia infections [72]. As described previously, the presence of chronic medical conditions may increase the risk of infection and associated complications. The number of adolescents with chronic medical conditions has increased in recent decades because of improving survival rates (e.g., for those with cystic fibrosis, congenital heart disease, cancer and spina bifida) and a rise in conditions such as HIV/AIDS and diabetes [73].

Current WHO and ACIP vaccine recommendations for adolescents with chronic medical conditions were reviewed recently by Hofstetter et al. [74]. These include HPV, Td/Tdap, meningococcal, pneumococcal and influenza vaccines [6,75] (Table 1). Conditions such as cancer, transplantation, HIV infection and systemic lupus erythematosus increase the risk of HPV infection and related complications [74]. Adolescents with severe chronic medical conditions also appear to be at increased risk of death following pertussis infection [76], supporting the ACIP recommendation of a shorter dosing interval for Tdap vaccination for this group [75]. The WHO and ACIP both recommend meningococcal vaccination of individuals with high-risk conditions beginning in early childhood [77,78], based on evidence that complement component deficiencies and asplenia increase susceptibility to meningococcal infection.

Certain chronic medical conditions (such as pulmonary, cardiac and neurological or neuromuscular diseases) place adolescents at increased risk of influenza-related complications, hospitalisation and death [74]. The ACIP now also recommends pneumococcal vaccination for children aged 6–18 years who have certain underlying conditions [75], supported by data showing that invasive pneumococcal disease is markedly elevated in this age group if haematologic malignancy, HIV/AIDS or sickle cell disease is present [79].

Where reported, vaccination coverage rates among adolescents with chronic medical conditions are variable but often low, lagging behind that for younger age groups [74]. For example, two studies of adolescent cancer survivors in the US found that only around one-third had received at least one HPV vaccine dose [80,81]. In England and Wales, the proportion of at-risk 15–19 year-olds who received pneumococcal vaccination ranged from less than 5% to approximately 65%, depending on the underlying medical condition [82]. Reasons for under-vaccination of individuals with

at-risk conditions in general also apply to the adolescent subgroup, such as misperceptions of the seriousness as well as risk of vaccine-preventable diseases and suboptimal coordination of primary and subspecialty care. Factors that uniquely affect adolescents with chronic medical conditions include the parental role in making vaccination decisions, often in the face of great complexity in terms of their child's condition and treatment regimens [74]. Logistical problems are frequently identified by parents as a reason for delayed or missed vaccinations [83]. Also, because of limited understanding of and comfort with adolescent health issues, healthcare providers may fail to provide preventive services to those with chronic medical conditions [84]. Interventions therefore need to be tailored to address the barriers to vaccination for these high-risk patients [74], including strategies to educate adolescents, parents and healthcare providers about vaccine-preventable diseases and improve provider-family communication [85].

5. Vaccination of older adults

Immunosenescence describes the decline in systemic immunity associated with aging involving both innate and adaptive immune responses [86]. Older adults are therefore typically more susceptible to vaccine-preventable diseases and disease can be more severe than in younger people (Box 3). This is a global problem and the WHO recommends annual vaccination of persons aged 65 years or older with seasonal inactivated influenza vaccine [6], while the ACIP recommends the following additional vaccines: Td/Tdap, herpes zoster and pneumococcal vaccines [87] (Table 1).

When vaccine strains closely match circulating influenza viruses, protection rates with inactivated influenza vaccines in individuals younger than 65 years are typically 70–90% [88]. In older people, the effectiveness of influenza vaccination is reduced, irrespective of setting, population and study design, though it remains the most efficacious public health tool available to protect elderly individuals against influenza. Attempts to improve effectiveness for people aged over 65 include the introduction in some countries of inactivated influenza formulations adjuvanted with an oil-in-water emulsion (MF59) or with increased antigen concentration as well as quadrivalent vaccines that potentially offer wider protection [88].

S. pneumoniae is a common cause of community-acquired pneumonia in people aged over 65 years, with higher morbidity and mortality than in younger individuals [89]. The ACIP has recently

Box 3

Vaccination of older adults.

Population	• Older adults
Reasons for increased risk of disease	• Decline in innate and adaptive (antibody and T-cell) immune responses (immunosenescence)
Diseases most commonly targeted by vaccination	• Influenza
	• Community-acquired pneumonia
	• Shingles (herpes zoster)
	• Tetanus
	• Diphtheria
	• Pertussis
Reported vaccine uptake rates ^a	• 50–70% for influenza, tetanus, pneumococcus
	• <25% for herpes zoster
Common barriers to vaccination	• Attitudes and beliefs regarding vaccination (especially negative attitudes)
	• Failure of healthcare provider to recommend vaccination
	• Lack of knowledge of vaccine safety and effectiveness
	• Perceived susceptibility to disease
Unmet vaccination needs	• Lack of awareness of national recommendations
	• Vaccines with improved immunogenicity, particularly influenza and pneumococcal vaccines
	• Vaccines against other leading causes of infection in older adults, such as <i>Staphylococcus aureus</i> , <i>Clostridium difficile</i>
	• Among active older adults, better consideration should be given to immunity to and risk of travel-associated diseases

^a In US.

recommended routine use of both pneumococcal conjugate and polysaccharide vaccines for adults in this age group [90], based on evidence from a randomised controlled trial with the 13-valent PCV [91]. Age-related declines in T-cell immunity are also predictive of an increased incidence and severity of herpes zoster [92,93]. The large Shingles Prevention Study showed efficacy with a concentrated live attenuated vaccine (*Zostavax*TM, Merck & Co, Inc., US) against herpes zoster and post-herpetic neuralgia in adults aged 60 years or older [94]. However, its efficacy against the occurrence of herpes zoster decreased with age, falling to 38% in adults aged 70 years or older [94], and protection ceased to be significant after eight years [95]. It is licensed for immunocompetent adults aged 60 years or older in many countries [96]. A new investigational herpes zoster subunit vaccine that combines a key virus surface glycoprotein with a T-cell boosting GSK proprietary adjuvant system (ASO1_B) showed efficacy of over 90% in 50–59, 60–69 and ≥ 70 years age groups in a randomised controlled trial [97,98].

Vaccination coverage rates for older adults show targets are often not being reached. The global target for influenza vaccination coverage is at least 75% for persons aged 65 years or older, but this has been achieved in few countries [99]. In the US, influenza vaccine coverage in the 2014–2015 season was 67% for this age group [100]; coverage for other vaccines in 2013 was 56% for tetanus, 60% for pneumococcus and (in adults aged 60 years or older) 24% for herpes zoster [101]. In developed economies, the most important factors related to vaccine uptake among older adults are people's attitudes and beliefs regarding vaccination, recommendations of healthcare providers, vaccine safety and effectiveness, and perceived susceptibility to disease [102]. Also, older adults may be unaware of the need for vaccination because of a lack of commitment from national health systems to fully implement vaccine recommendations [103]. Uptake rates may be improved with public education campaigns, by introducing appropriate operational frameworks and by introducing new vaccine formulations that enhance immune responses with additional adjuvants, increased dose or other changes in vaccine composition [104]. In particular, the development of more immunogenic influenza and pneumococcal vaccines of broader specificity is actively being pursued. In developing countries, the need for better vaccination coverage of aging populations is recognised [105] and influenza vaccine effectiveness appears to be similar to that reported in high-income countries [106], but developing countries struggle to implement WHO recommendations on influenza vaccination [107].

6. Conclusions

To achieve one of the strategic objectives set by the GVAP – to equitably extend the benefits of immunisation to all people – every eligible individual should be immunised with all appropriate vaccines [1]. The Pan American Health Organization has also promoted a complete transition from child to family immunisation programmes, in which countries incorporate into their national schedules appropriate vaccines for all family members throughout their lives [108]. Consistent with these objectives, target populations for vaccination encompass special populations who are especially vulnerable to vaccine-preventable infections, including pregnant women, preterm infants, people with chronic and immune-compromising conditions and older adults.

It is clear that special groups are often under-vaccinated due to a variety of reasons, including lack of awareness of vaccine-preventable diseases and healthcare systems that fail to ensure such patients receive recommended vaccines. This is particularly applicable to conditions where patients receive care from multiple subspecialists and little or no care from primary care providers. To reach the unvaccinated, there is a need for greater awareness

among patients, caregivers and healthcare providers of the value of vaccines and vulnerability to vaccine-preventable diseases at different ages or in association with different health conditions. There is an obvious need for improved understanding of basic aspects of vaccines among special populations and the healthcare professionals involved in their care, including the specific immunopathogenesis of target diseases, the epidemiology of vaccine-preventable infections, effects of vaccines and optimal timing of recommended immunisation. There is also evidence that healthcare providers' attitudes towards vaccination are among the most important influences on the decision to vaccinate but that physicians' adherence to recommendations needs to be improved. The fact that vaccination rates among at-risk populations remain low despite recommendations indicates a continuing failure to provide appropriate standards of care. Reversing this is likely to require a broad range of interventions. Financial incentives, patient reminders and patient recall systems all appear to improve vaccination rates and can be more readily implemented in high income country settings [109,110]. The data from low and middle income countries are less robust but suggest that patient reminders and immunisation outreach could work in these settings also [111–113].

Trademarks

Synflorix is a trademark of the GSK group of companies. *Zostavax* is a trademark of Merck & Co., Inc.

Contributorship

All authors were involved in the development of this manuscript and gave final approval before submission.

Disclosures

MD and RS-O are employees of the GSK group of companies and report ownership of restricted shares in the GSK group of companies. JIS and SLR report no conflicts of interest. LRS reports consulting fees from Vical, Genocoea, and Janssen Pharmaceuticals, and is a member of the scientific advisory board of Abivax. AMH reports grants from Pfizer Medical Education Group and from Merck Sharp & Dohme Corp. ALC reports consultancy fees to his institution, unrelated to the present work, from the GSK group of companies.

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